

Fractal Information Theory Project

FIN ST327–ST341

Certified Coexistence, Mediator Rank,
and Quantitative Observer Depth

Analytical, interval, and bounded computational research report

Krzysztof Żuchowski

Independent Researcher — Fractal Information Theory Project
ORCID: 0009-0002-0909-3613

August 16, 2026

Release 10.79

Repository snapshot. Audited tracked base: `bb540a767e210a5a3bf816e4868e7bcefbac3872`. The working tree contains uncommitted research artifacts; the release SHA-256 ledger identifies the files.

License. CC BY 4.0. **Language.** English. **Resource type.** Preprint and executable reproducibility package.

Abstract

Programs ST327–ST341 continue the local mathematical program after Release 10.78. The main advance is theorem-level local phase coexistence. Outward endpoint boxes and one parametric Krawczyk tube prove a unique positive reflection-even stationary root for every $g \in [2.900, 2.905]$. Its objective is strictly positive at $g = 2.900$ and strictly negative at $g = 2.905$, so at least one energy-equality crossing with the uniform state exists in that interval. This is not yet a global transition: the exact 11-dimensional dual is compact on the relevant sublevel, but a complete tensor cover exceeds the frozen budget by orders of magnitude.

The localized branch has a well-conditioned augmented fold candidate at $g \approx 2.664908051$, still numerical. Lower-rank conservative mediators are classified exactly: the optimal rank- m Schur term retains the m largest strict eigenmodes, with Eckart–Young errors equal to the discarded spectral norms. A strict-invariant quartic $\langle q, Aq \rangle^2$ supplies an admissible stabilizer shape, but its positive coefficient is not sourced.

On the observer side, binary child-resolution instruments acquire an exact Le Cam depth: for errors $e_f \leq e_c < 1/2$, the finer experiment has zero deficiency toward the coarser, whereas reverse deficiency is $e_c - e_f$. Finite counts impose an $N^{-1/2}$ recovery floor. Tree scale cocycles are gauge, but a cyclic refinement carries the gauge-invariant holonomy $H = g_{01}g_{12}/g_{02}$. Approximate refinement preserves relative clocks within explicit Weyl intervals. A declared correlated Markov branch process admits rate–distortion below the arbitrary-history rate, showing precisely what fractal compression can mean after a state class and loss function are supplied.

No strict source for active coupling or stabilizer coefficient, global physical vacuum, selector closure, absolute unit, time arrow, Planck scale, spacetime, laboratory evidence, legacy-role transfer, Standard Model, gravity, L_{total} , or Theory-of-Everything closure is claimed.

Contents

1	Scope and proof discipline	1
2	Executive ledger	1
3	ST327: a certified local coexistence transition	1
4	ST328–ST331: fold, globality, mediator rank, and stabilization	2
5	ST332–ST334: tails, branch tubes, and multi-time infrared design	3
6	ST335–ST337: cyclic calibration and quantitative observer depth	4
7	ST338–ST340: controlled compression, approximate clocks, and dimensions	4
8	Falsification ledger and current interpretation	5
9	Recommended programs ST342–ST356	6
10	Reproducibility and conclusion	7

1 Scope and proof discipline

The strict object remains

$$A = sI - W, \quad W_{xx} = 0, \quad W_{xy} = K_{\text{strict}}(d(x, y)) \ (x \neq y),$$

with $A1 = 0$ and $A \succeq 0$. The legacy kernel remains an intermediate bridge object; no legacy role is transferred. The internal ontology remains informational nadsoliton $\rightarrow$ light $\rightarrow$ matter $\rightarrow$ emergent observer, without a separate informational layer below the nadsoliton. It is not used as a proof premise.

[PROVEN] marks exact analysis or outward finite certification. **[STRONG NUMERICAL EVIDENCE]** marks reproducible floating computation. **[PARTIAL]** means a required continuous or global step remains open. **[BLOCKED]** records absent external evidence.

2 Executive ledger

Pro-gram	Status	Result
ST327	[PROVEN]	A parametric root tube and opposite endpoint energy signs prove local coexistence crossing in $[2.900, 2.905]$.
ST328	[STRONG NUMERICAL EVIDENCE]	Simple-fold candidate at $g \approx 2.664908051$; no 17-dimensional interval certificate.
ST329	[PROVEN] + [PARTIAL]	Relevant dual sublevel is compact, but complete 11D tensor covering is outside the frozen budget.
ST330	[PROVEN]	Exact optimal rank/error tradeoff for lower-rank conservative mediators.
ST331	[PROVEN] + [PARTIAL]	Strict A supplies a quartic invariant shape; its positive coefficient remains unsourced.
ST332	[PROVEN] + [PARTIAL]	Projective observables have geometric tails; the weighted Chernoff derivative tail remains open.
ST333	[PARTIAL]	Only 26 of 40 isotropic overlapping slab attempts pass; continuous branch loss is not inferred.
ST334	[PROVEN] + [STRONG NUMERICAL EVIDENCE]	Finite multi-time IR design is an LP with dual threshold-band structure; grid solution is not a continuum theorem.
ST335	[PROVEN]	Minimal cyclic refinement has gauge-invariant scale holonomy H .
ST336	[PROVEN]	Exact Le Cam deficiency for binary partial child-resolution instruments.
ST337	[PROVEN]	Finite-count scalar fiber recovery has an $N^{-1/2}$ minimax floor.
ST338	[STRONG NUMERICAL EVIDENCE]	Block Markov rate–distortion demonstrates controlled correlated-history compression.
ST339	[PROVEN]	Approximate intertwining yields explicit Weyl bounds for relative clock ratios.
ST340	[PROVEN]	Dimensional rank and raw- A unit incompatibility are separated from operational identifiability.
ST341	[BLOCKED]	No independent raw-event packet exists.

3 ST327: a certified local coexistence transition

For a reflection-even probability p represented by seven orbit values q_i , stationarity at supplied coupling g is

$$\log(12q_i) + 1 - g(Bq)_i + \lambda = 0, \quad q_0 + 2 \sum_{i=1}^5 q_i + q_6 = 1.$$

At $g = 2.900$, the outward objective interval is

$$[0.0016740014, 0.0016751191],$$

whereas at $g = 2.905$ it is

$$[-0.0016823923, -0.0016812705].$$

Both endpoint roots lie in unique Krawczyk boxes. More importantly, one coordinatewise parametric box contains a unique positive root for every g in the complete interval; its minimum inclusion margin is 6.7389×10^{-5} .

Theorem 3.1 (certified local coexistence). *The positive reflection-even localized branch exists continuously throughout $g \in [2.900, 2.905]$ and has at least one point at which its objective equals that of the uniform state.*

This is the first theorem-level transition statement for the supplied negative-information model. It is local coexistence, not a global phase transition: another state may have lower energy, and the crossing need not be unique. The coefficient g remains supplied.

4 ST328–ST331: fold, globality, mediator rank, and stabilization

The augmented equations

$$F(x, g) = 0, \quad D_x F(x, g)v = 0, \quad \|v\| = 1$$

have a floating root at

$$g_{\text{fold}} \approx 2.66490805100556.$$

The residual is below 5×10^{-14} , the stationary Jacobian has one numerical null direction, and the augmented Jacobian has minimum singular value about 0.255. This is strong evidence for a simple spinodal, but no outward 17-dimensional box was obtained.

ST329 proves a useful compactness result for the exact dual. On the mean-zero subspace,

$$\frac{1}{2g} \langle h, A^+ h \rangle - \log \left(\frac{1}{12} \sum_i e^{h_i} \right) \geq \frac{\|h\|^2}{2g\lambda_{\max}} - \|h\|.$$

Hence every negative sublevel lies inside $\|h\| \leq 2g\lambda_{\max} \approx 18.7375$. The problem is compact, but a mere 16-cell partition per axis already contains $16^{11} = 17,592,186,044,416$ boxes. The frozen million-box budget cannot support a tensor cover. This is a method no-go, not a globality verdict.

For a mediator of dimension m , the induced Schur matrix has rank at most m .

Theorem 4.1 (optimal lower-rank mediator). *Among positive semidefinite matrices of rank at most m , the best approximation to gA in either Frobenius or operator norm retains the m largest eigenvalues and modes of gA . The errors are respectively*

$$\left(\sum_{j>m} \sigma_j^2 \right)^{1/2}, \quad \sigma_{m+1}.$$

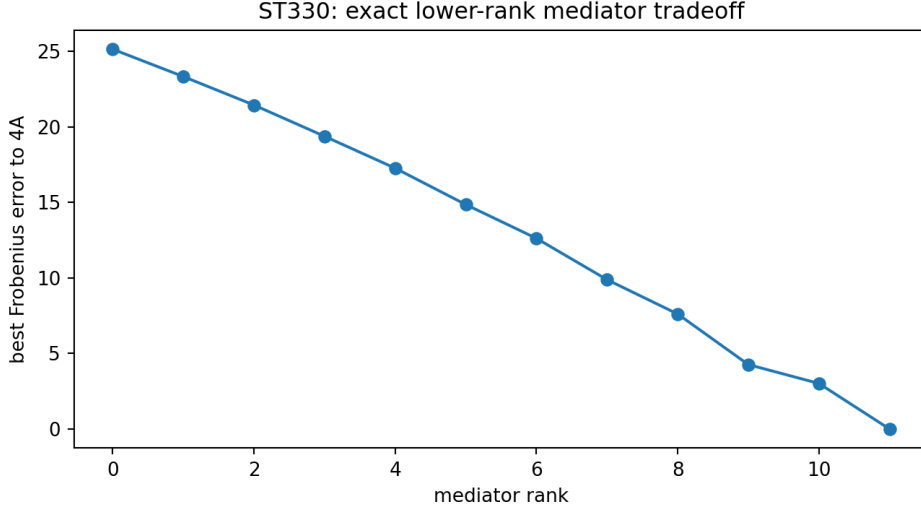


Figure 1: Exact cost of reducing the conservative mediator below the full rank 11 required for gA .

Lower-rank mediation therefore selects stiff modes rather than reproducing the full strict geometry. FIN currently supplies no rule selecting the rank or coupling.

An admissible stabilizer is

$$V_{g,c}(p) = D(p||u) - \frac{g}{2}Q + \frac{c}{4}Q^2, \quad Q = \langle p - u, A(p - u) \rangle.$$

The shape is D_{12} -invariant and constructed only from strict A ; $c > 0$ stabilizes an unconstrained mean-zero carrier. But the probability simplex is already compact, and A does not choose the coefficient or its sign. This is a completion template, not a sourced law.

5 ST332–ST334: tails, branch tubes, and multi-time infrared design

The exact filter contraction $\rho = 5/7$ implies that every projective-Lipschitz observable with constant L has future dependence tail at most

$$\frac{L\rho^n}{1-\rho}.$$

For the frozen one-step log-weight oscillation $L \leq \log 49$, this drops from about 1.29 at depth 7 to 2.78×10^{-11} at depth 80. The full Chernoff derivative also differentiates multiplicative weights; no uniform normalized derivative Lipschitz constant was certified. Belief forgetting is proved, optimizer promotion is not.

ST333 tests 40 wider isotropic boxes intended to overlap the ST298 sections. Twenty-six pass and fourteen fail. The first failures are recorded in the packet. This falsifies that simple enclosure strategy, not the existence of the branch. A true parameterized tube needs separate longitudinal and transverse widths.

For several observer times t_j , the soft-IR problem is the linear program

$$\max_{0 \leq q \leq 1} z, \text{quad} \int d_{t_j}(\mu)q(\mu)d\log \mu \geq z, \text{quad} \int h(\mu)q(\mu)d\log \mu \leq B.$$

LP duality proves a threshold rule for one weighted combination of the active time kernels minus the heat price. Unlike the single-time bathtub order, several bands can occur. The 500-point, three-time grid gives minimax value 1.2763170 under budget 3, but remains a finite discretization.

6 ST335–ST337: cyclic calibration and quantitative observer depth

Tree cocycles are gauge-trivial. The smallest nontrivial refinement diagram is a triangle. Under vertex calibration changes, the edge rates change but

$$H = \frac{g_{01}g_{12}}{g_{02}}$$

does not. $H = 1$ is path independence; $H \neq 1$ is relative calibration curvature. This is a complete one-cycle classification, but FIN has not supplied a physical cyclic refinement or a value of H .

The qualitative Blackwell order can be made quantitative. Let a child-resolving instrument be a binary symmetric channel with error e .

Theorem 6.1 (exact binary observer depth). *For $0 \leq e_f \leq e_c < 1/2$,*

$$\delta(\text{BSC}(e_f), \text{BSC}(e_c)) = 0, \quad \delta(\text{BSC}(e_c), \text{BSC}(e_f)) = e_c - e_f.$$

The finer instrument simulates the coarser by adding noise. The reverse direction cannot increase total-variation separation. This creates a continuous operational layer depth between exact resolution and blindness. It is an instrument theorem, not a theory of consciousness or proof of a physical FIN layer.

Ideal identifiability also does not remove sampling cost. Le Cam’s two-point method gives a scalar child-probability minimax floor of order $N^{-1/2}$. Extending to 78 free symmetric fiber parameters requires at least order $78/\epsilon^2$ effective independent counts before confidence and conditioning factors. No local code can replace those events.

7 ST338–ST340: controlled compression, approximate clocks, and dimensions

ST326 proved that arbitrary binary branch histories require one bit per layer. Compression becomes possible only after defining a state class. ST338 chooses a stationary binary Markov class with flip probability 0.1 and normalized Hamming distortion. Its asymptotic lossless entropy rate is

$$h_2(0.1) \approx 0.468996 \text{ bits per layer},$$

already below the arbitrary-history rate. An eight-layer Blahut–Arimoto computation gives the displayed finite-block rate–distortion curve.

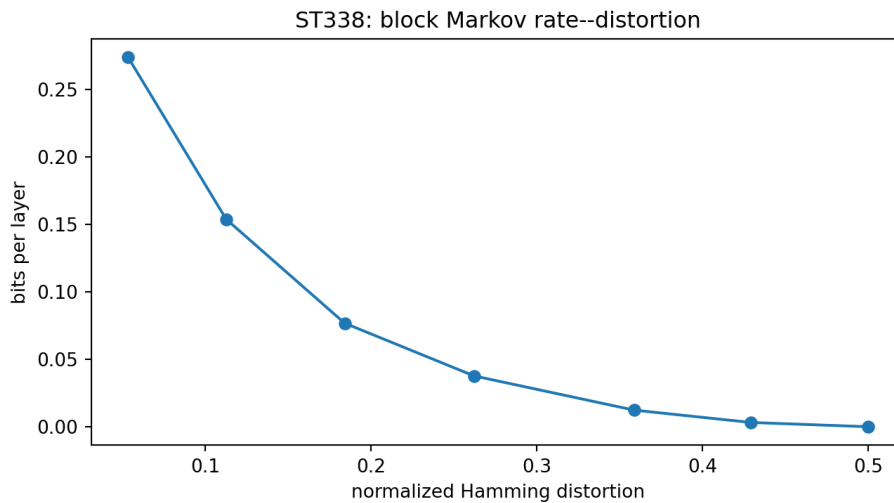


Figure 2: Controlled compression after supplying a correlated source class and a distortion measure.

This is the rigorous form of the useful fractal-compression intuition: a short rule plus correlations and allowed distortion can reduce record cost. The Markov law, its parameter, and the distortion are not derived from FIN.

Relative clocks are stable when exact intertwining is weakened. If the compressed refined operator differs from A by operator norm at most $\epsilon < \lambda_1$, Weyl's theorem gives

$$\frac{\lambda_j - \epsilon}{\lambda_i + \epsilon} \leq \frac{\tilde{\lambda}_j}{\tilde{\lambda}_i} \leq \frac{\lambda_j + \epsilon}{\lambda_i - \epsilon}.$$

Wave-clock ratios obey the square roots of these bounds. This controls relative drift; it does not produce seconds or a time arrow.

Finally, dimensional relations and channel identification are separate obligations. In basis (L, T, E) , a speed row $(1, -1, 0)$ and action row $(0, 1, 1)$ have rank two. Adding an independent diffusion row can reach rank three. Repeating ET does not. More basically, one raw A cannot simultaneously carry dimension T^{-1} for unitary/heat evolution and T^{-2} for wave evolution. Distinct sector conversion maps are necessary and must be operationally calibrated.

8 Falsification ledger and current interpretation

Claim	Test	Surviving statement
The sampled crossing was enough	Parametric root tube and interval energies	Now repaired: at least one local crossing in $[2.900, 2.905]$ is proved.
The crossing is a global transition	Compact dual plus cover audit	Still unproved; naive 11D tensor certification is infeasible under the frozen budget.
The spinodal is proved	Augmented fold system	Only strong numerical evidence at $g \approx 2.664908051$.
Small mediator reproduces strict gain	Eckart–Young classification	False exactly below rank 11; lower ranks approximate selected stiff modes.
Strict symmetry supplies stabilization completely	Quartic invariant audit	It supplies a shape, not a positive coefficient.
Belief contraction closes Chernoff asymptotics	Weighted derivative audit	False so far; weight derivatives require another uniform constant.
All scale cocycles are gauge	Minimal cycle	False on cyclic diagrams; holonomy is gauge invariant.
Observer depth is merely qualitative	Binary deficiency theorem	False in the declared family; exact quantitative depth is $e_c - e_f$.
Fractal compression is impossible	Markov rate–distortion	False for restricted correlated sources; arbitrary-history lower bounds remain intact.
Relative clocks collapse under small error	Weyl bounds	False; they vary continuously inside explicit intervals.
Local work supplies evidence	Independent-record gate	False; ST341 remains blocked.

The deepest surviving interpretation is:

With a supplied signed interaction, FIN now possesses a theorem-level localized branch and a certified local coexistence crossing. What remains missing is global variational closure and a strict source for the coupling and stabilization coefficient. Refinement supports a quantitative hierarchy of accessible information, gauge-invariant calibration holonomy on

cycles, and robust relative clocks. Fractal compression is meaningful only relative to a source class and tolerated distortion. These are precise mathematical bridges, but none selects SI units, an apparatus, or our observed universe without additional operational and empirical structure.

9 Recommended programs ST342–ST356

- ST342.** Narrow the certified coexistence bracket using interval bisection on the parametric root tube.
- ST343.** Build a full 17-dimensional interval Krawczyk certificate for the fold candidate.
- ST344.** Reduce the exact global dual by D_{12} orbit types and derive a symmetry-safe lower envelope before any branch-and-bound.
- ST345.** Prove or refute that every global minimizer at $g = 4$ is reflection-equivalent to the certified localized orbit.
- ST346.** Couple optimal rank- m mediators to the simplex and determine which modal truncation first produces localization.
- ST347.** Classify all degree-four and degree-six strict invariants that can stabilize the carrier and expose every free coefficient.
- ST348.** Derive the missing normalized Chernoff derivative Lipschitz constant or construct a counterexample.
- ST349.** Replace isotropic branch boxes by anisotropic interval implicit-function slabs.
- ST350.** Certify the continuum multi-time IR dual with finitely supported dual measures or prove that support must grow.
- ST351.** Classify scale holonomy on two-cycle refinement complexes and its independent cohomology rank.
- ST352.** Extend Le Cam observer depth from BSC instruments to the 12-child circulant confusion family.
- ST353.** Derive sharp finite-count lower and upper bounds for recovering a spectrally restricted fiber B .
- ST354.** Estimate rate–distortion for a source law generated by a declared FIN transition kernel, keeping the law explicitly conditional.
- ST355.** Propagate approximate-clock bounds through multiple refinement levels and separate accumulated drift from gauge calibration.
- ST356.** Resume empirical analysis only after independently registered, hashed raw events and separated custody roles exist.

The highest priorities are ST343 and ST344: they target the remaining local fold certificate and the global variational obstruction. ST352 is the strongest continuation of the internal observer program because it moves from a binary toy family to the actual 12-state symmetry without claiming an apparatus.

10 Reproducibility and conclusion

The executable source `fin_st327_st341_research.py` writes fifteen packets, an aggregate JSON, a CSV ledger, and two figures. The independent suite `test_fin_st327_st341.py` performs sixteen live checks of packet hashes, crossing signs, parametric inclusion, the fold residual, cover-budget stop, mediator error monotonicity, stabilizer status, projective tails, slab failure, LP feasibility, holonomy invariance, deficiency, finite-count scaling, rate–distortion monotonicity, Weyl containment, dimensional rank, and the evidence stop. All sixteen tests pass.

Release 10.79 closes one important gap: local coexistence is now a theorem rather than a plot. It also shows why this is not yet physical closure. Globality, signed-source origin, stabilization strength, dimensional calibration, apparatus, and independent events remain separate obligations. The observer and compression ideas survive in more rigorous forms: deficiency rather than pictorial depth, holonomy rather than arbitrary scale rate, and rate–distortion rather than cost-free storage of arbitrary histories.